

ANDRY HUANG

Senior Fullstack & AI Engineer (.NET | Next.js | Node.js | RAG / AI Agents)

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PROFESSIONAL SUMMARY

Senior Fullstack & AI Engineer with 7+ years of experience. Spent 3+ years remote at MRI Software (Singapore) building enterprise leasing and tenant-management SaaS, including production RAG features on OpenAI and pgvector. Founded Qualiv, an AI recruitment SaaS, and built ERP, POS, and CMS systems for businesses in Surabaya. Core stack: .NET Core, Next.js, Node.js, NestJS, SQL tuning, and LLM/RAG.

TECHNICAL SKILLS

- **Backend & Cloud Architecture:** .NET Core, C#, Node.js, NestJS, Laravel, Java Spring Boot, Python (FastAPI), REST & GraphQL APIs, SaaS Architecture
- **AI Engineering & Automation:** RAG Architecture, OpenAI GPT-4o, Anthropic Claude API, Model Context Protocol (MCP), Autonomous AI Agents, pgvector, BullMQ, Prompt Engineering
- **Frontend & Mobile:** Next.js, React, TypeScript, JavaScript, Tailwind CSS, Flutter, Android Native, Livewire, Alpine.js
- **Databases & Storage:** PostgreSQL, SQL Server, MySQL, Oracle DB, Redis, Elasticsearch, AWS S3, Azure Blob, Cloudflare R2
- **DevOps, CI/CD & Security:** Docker, GitHub Actions, Azure DevOps, GitLab CI, Linux Administration, Nginx, Fail2ban, SonarQube, SSH Security
- **Testing & Integrations:** Playwright, Selenium, Midtrans Payment Gateway, RajaOngkir, RFID Hardware, Leaflet GIS

WORK EXPERIENCE

Fullstack Developer | MRI Software (formerly Anacle Systems Ltd) (Singapore, Remote) | Nov 2022 - Mar 2026

- Built features for Anacle Simplicity and the SMRT Tenant Management System through go-live: mall tender submissions, space leasing, tenant applications, and daily operations. Stack: .NET Core, .NET Framework 4.8, Next.js, NestJS.
- Shipped production RAG pipelines on the OpenAI API and PostgreSQL (pgvector) for semantic document search and automated executive reports.
- Wrote a microservice that migrated 100,000+ files to SharePoint (Graph API), AWS S3, and Azure Blob Storage, with RBAC and audit logging.
- Refactored legacy stored procedures and wrote regression tests in Selenium and Playwright for core SaaS modules.
- Led the SonarQube rollout across enterprise repositories, fixed the security vulnerabilities it flagged, and set CI/CD quality gates that block deployments with new vulnerabilities.

Freelance Fullstack & AI Engineer | Self-employed (Surabaya, Remote) | 2021 - Present

- Build AI products end to end, from architecture to production: Qualiv, an AI recruitment SaaS, and an AI tire-recommendation assistant for PT Nagamasban.
- Deliver client systems: the Wonderful Works website and CMS, and Laravel ERP/POS systems for tire, jewelry (RFID), yarn, and grocery retailers.

Fullstack Developer | Software House (Surabaya, Indonesia) | Mar 2022 - Nov 2022

- Maintained a legacy enterprise Tool Management System (TMS) in Java, Spring Boot, and Oracle. Led a Java version upgrade, a large refactor, and a code-review process, and optimized complex SQL queries.
- Tracked down the edge-case business-logic bugs that were corrupting data, patched them, and repaired 10,000+ corrupted records.
- Maintained self-hosted Git servers and set up Elasticsearch clusters for log indexing.
- Traced recurring high-priority incidents to backend bottlenecks and fixed them, cutting incident tickets quarter over quarter.

Full-Stack Developer | PT Pabrik Kertas Tjiwi Kimia Tbk (Surabaya / Mojokerto, Indonesia) | Sep 2019 - Feb 2022

- Built manufacturing dashboards and reports for pulp & paper production in .NET Core, C#, ASP Classic, and SQL Server. Query tuning cut long-running reports from minutes to seconds.
- Automated plant reporting with a WhatsApp bot (Python, Java, Selenium) that sends scheduled dashboard snapshots of OEE and HWT production metrics.
- Shipped a native Android barcode scanner app for warehouse stock-in and stock-out.
- Developed internal apps, including a company-wide daily COVID-19 health check-in and a digital company raffle draw.

FEATURED PROJECTS

- **Qualiv | AI Recruitment SaaS Platform** (Founder & Lead Architect): Multi-tenant recruitment SaaS, built from scratch. It parses CVs (PDF/DOCX), runs logic tests and async video interviews, and scores each candidate against the job's requirements. Tech Stack: Next.js, NestJS, OpenAI GPT-4o, PostgreSQL, Prisma, BullMQ, Redis Queue, Cloudflare R2, Midtrans
- **PT Nagamasban | AI Sales Assistant & Catalog** (Lead AI Architect): Recommendation and sales-training tool for a Surabaya tire distributor. Matches tires to a customer's vehicle and budget using OpenAI embeddings and semantic search. Tech Stack: Next.js, NestJS, OpenAI Embeddings, Semantic Search, Tailwind CSS
- **Wonderful Works | Architecture & Contractor Platform** (Fullstack Architect): Company website and CMS for a Surabaya architecture and contracting firm. Staff can edit pages in a drag-and-drop composer and manage bilingual (ID/EN) content, project photos, and users from an admin panel with JWT auth. JSON-LD for SEO; deploys through Docker and Jenkins. Tech Stack: Next.js 16, React 19, Tailwind CSS v4, Prisma, PostgreSQL, Puck Visual Editor, Docker, Jenkins
- **Retail ERP & POS Systems** (Fullstack Architect): Custom systems for four retail businesses: Cahaya Terang Tire ERP (double-entry A/R and A/P ledger), Wijaya Mas Jewelry POS (RFID stock tagging), Knit and Cro e-commerce (Midtrans, RajaOngkir), and Tiga Putra Grocery POS (FIFO expiry tracking). Tech Stack: Laravel, Livewire, MySQL, RFID Integration, Payment Gateway

EDUCATION

Bachelor of Science in Computer Science (S1 Teknik Informatika) | University of Surabaya (UBAYA), Indonesia (2015 - 2019) | GPA: 3.60 / 4.00 (Cum Laude)

CERTIFICATIONS

- **Anthropic AI Certifications Suite (9 Certifications)** (Anthropic, 2026): **Building with Claude API** | **Claude Code in Action** | **Intro to MCP** | **MCP Advanced** | **Agent Skills** | **Subagents** | **AI Capabilities & Limitations** | **Claude Cowork** | **Claude 101**
- **AI Agents with Model Context Protocol: Vanderbilt University** (**Coursera Verification**, 2026). Hands-on course covering agent design, tool use, and building MCP servers and clients.
- **Software Architecture & Design of Modern Large Scale Systems**: **Udemy** (**Udemy Verification**, 2026). Covers microservices, caching, and fault-tolerant design for high-throughput systems.

LANGUAGES

- English (Professional Working Proficiency), Indonesian (Native Proficiency)